

A Wearable Haptic Interface for Assisting Blind and Visually Impaired Students in Learning Algebraic Equations

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ABSTRACT

BVI (blind and visually impaired) persons do not have the same degree of access to concepts and procedures in learning algebraic equations as sighted persons. As tactile resources are uncommon, most rely on auditory-based methods that place heavy stress on working memory. We propose a novel approach that leverages haptic technology to communicate mathematical symbols via vibrotactile patterns; this design addresses open challenges found in existing resources for BVI people and explores the potential utility of this medium for all users. We conducted three separate studies to evaluate the feasibility of our gloves that included BVI participants. The outcomes of our studies yielded insightful design discussions, including a blind participant who found the device immediately intuitive and exciting in potential.

CCS CONCEPTS

• **Human-centered computing** → *Interface design prototyping; Accessibility technologies; Haptic devices.*

KEYWORDS

haptic feedback, tactile feedback, visually impaired, wearables, haptic glove, math notation

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1 INTRODUCTION

Algebra is a foundational topic within the field of mathematics, providing individuals with a language to analyze, communicate, and model quantitative relationships and real-world data. For students, mastery of this topic is essential not only for learning advanced fields within almost every subject of schooling but also for obtaining a desirable job within today's job market [13, 23, 47]. To achieve this fluency, students must master working with algebraic equations, which are mathematical statements that are expressed as an equality comparison of two mathematical expressions. However, for blind and visually impaired (BVI) students, this can be quite challenging, as algebraic equations are communicated visually through sequences of specific symbols (e.g., letters, numbers, operators, brackets). In practice, BVI students need to commit equations and the steps they take to working memory, inducing significant cognitive load and leading to BVI students solving algebra problems much slower than sighted students who have continuous and instantaneous access to equations and the steps they have taken afterwards [17, 22]. Furthermore, spatial reasoning ability associated with algebraic equations often relies on visual cues and can further limit comprehension of mathematical concepts for BVI students [41]. As a result, achievement in mathematics among BVI individuals is notably lower: 75% of BVI students are more than a full grade level behind sighted students and 20% of BVI students five or more grade levels behind [14].

Conventional resources for communicating mathematical concepts such as algebraic expressions to BVI students are specialized braille alphabets which are traditionally written on embossed paper or using electronic braille displays. Text-to-speech (TTS) systems, where screen reader software applications convert images and learning contents into audio or braille, have also seen some use [3]. Jackson et al. [16] observed that Braille displays were more efficient for students compared to speech synthesizers due to the ability of the user to access and review more detailed information from the math problem. Alnfai et al. [3] had success with using gestures to distinguish the status of the Braille dot for every character—which

includes a tapping gesture to insert the Braille code and swiping gestures for editing—using a BrailleEnter keyboard.

One means by which algebraic expressions could also be communicated to BVI students is haptics. Prior work has shown that messages conveyed via vibrations are fast, reliable, and inconspicuous [40]. Furthermore, people can notice and differentiate between multiple vibrations, allowing for the communication of varied and complex messages [12]. While several studies have developed haptic systems specifically for BVI individuals [10, 19, 20], these studies have primarily been tangible interfaces designed to convey broad concepts rather than a means of conveying more specific messages. Thus we ground our work in the premise that haptics is both a logical and under-explored medium for communicating and interacting with algebraic expressions.

Specifically, in this work we present a preliminary haptic interface for increasing accessibility towards the mathematical notation of algebraic equations. Our interface consists of haptic gloves with vibration motors and buttons placed on the fingers to facilitate interaction with algebraic equations. We conducted a pilot study with this interface, seeking to learn whether BVI students can understand, navigate, and communicate basic mathematical symbols through vibrotactile sensors and actuators placed on an individual's hands. Specifically we aim to answer the following research questions:

- What is the ideal configuration of vibrotactile communication frequency, location, pattern, and duration to allow a user to recognize tactons representing basic mathematical symbols most accurately?
- How much mental effort is required to use a hand-centric haptic interface for basic mathematical symbols?

2 RELATED WORK

2.1 Haptic Communication

Haptic interfaces have long been used as a means of conveying information to individuals. They represent an alternative modality to the more commonly used senses of vision and audio, and studies have shown that users can accurately perceive haptic messages while using their other senses unimpeded [26]. In particular, researchers have noted that simple messages conveyed by singular tactons are the most easily comprehended; however, more complex messages conveyed by an array of tactons can be effectively communicated (though commonly only with training) [29, 48]. Tacton placement has varied widely in the literature; though several studies have concluded the hands and wrists represent the most effective locations for placing tactons that can be discerned [8, 11, 18, 28, 29, 36–39]. To leverage the effectiveness of this body location, many researchers have utilized smartwatches [6, 7, 25, 44], rings [40], and gloves [26–28].

Researchers have used vibrotactile systems to convey a variety of messages. Several studies have sought to convey rather simple messages. Wang et al. [48] conveyed a simple on/off when users received a notification, while Lylykangas [29] only conveyed messages for accelerate, keep speed constant, and decelerate. Other studies have sought to use vibrations as a means of conveying mobile notifications and time [8, 43, 49]. More commonly studies have sought to convey alphanumeric characters and words such

as in work by Luzhnica et al. [26–28], Liao et al. [25], and Blum et al. [6, 7].

2.2 Haptic Technology for BVI Individuals

Given its effectiveness at conveying information, haptics has been explored specifically as a means of improving accessibility for BVI individuals by a number of researchers. Researchers have explored its utility in a variety of domains such as conveying music [46, 50], driving [42], exergaming [31, 32], and creating digital braille interfaces [34, 35]. Notably, there have been a number of works that have specifically focused on conveying mathematical concepts. LMath [4] makes it possible for visually impaired users to more naturally interact with formulas via a haptic and audio interface. MathMelodies [1] combines engagement and accessibility by having an associated sound to make the interface accessible to visually impaired children. This application was later expanded into MathMelodies 2 and developed in React Native to make it accessible to people with Android devices [9]. These interfaces are not exclusively for education, e.g., Gutschmidt [15] developed a version of Sudoku that could be played by visually impaired individuals.

Beyond just numbers, haptics have also served as natural choice for communicating graphical concepts to students. Several studies have looked at conveying graphs and geometric concepts to students [2, 5, 30]. Haptic interfaces have also enabled students to interact with an explore scientific models, e.g., a plant cell or an atom, and gain a deeper understanding of the concepts by making them physically interactable [33]. This type of experiential learning allows the instructor to embed direct application of the concept into the lesson, e.g., learning about the concept of surface area by interacting with unfoldable cubes of varying sizes [10].

3 DESIGN

This work leverages the Nemeth Braille code for Mathematics in designing tactons for basic mathematical notation [24]. Nemeth braille is a specialized form of braille that maps mathematical and scientific notation to the standard six dot braille cell. For the purposes of this study we used the following symbols: 0–9, X , $+$, $-$, $=$, $\times$, $\div$. We provide the Nemeth braille characters for these symbols in Figure 1. This design choice serves to make the tacton design less arbitrary and leaves the pinky and thumb free for interaction functions.

We developed a pair of gloves with embedded vibration motors to facilitate tactile communication as the hand provides several locations for vibrotactile recognition [45]. Specifically, our system consisted of six vibration motors placed on the back of the index, middle, and ring fingers of both hands, as well as two additional vibration motors placed between the thumbs to represent the spaces. To input symbols, we placed buttons on the pads of the index, middle, and ring fingers of both hands. We conducted two studies to finalize the design of the glove. In the perception study, we sought to understand the optimal placement and parameters of the vibration motors on each of the fingers. In the navigation study, we sought to understand the best means of enabling navigation through the equations. These studies are described in the following subsections.

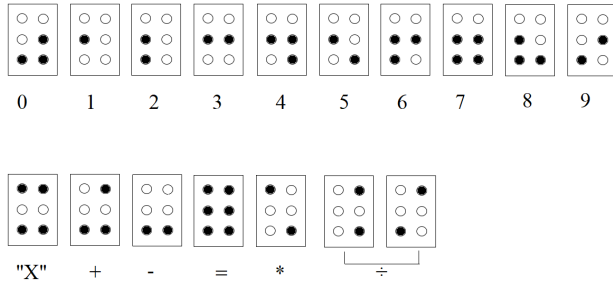
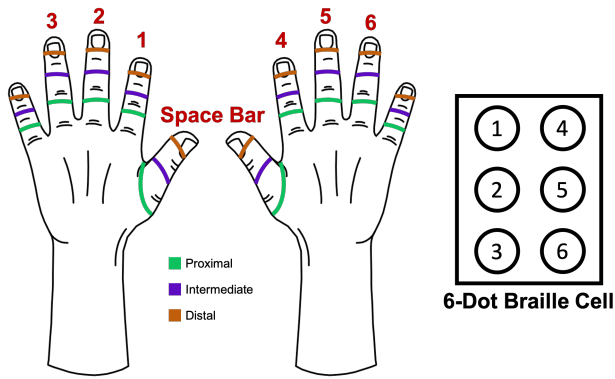


Figure 1: The six-dot Nemeth braille characters for the supported symbols



(a) Options for vibration motor placement associated six-dot braille cell format



(b) Posterior view of the haptic glove interface

Figure 2: Design considerations for the development of a wearable braille interface

3.1 Perception Study

We conducted a user study to determine both the optimal placement of, frequency of, and overlap between the vibration motors to maximize recognizability and minimize mental effort. Specifically,

we varied the location on the thumb, index, middle, and ring fingers (i.e., distal, intermediate, proximal) (shown in Figure 2a, frequency of the vibrations (i.e., 10 Hz, 20 Hz, 30 Hz), and the amount of overlap between sequential vibrational stimuli (i.e., complete overlap, partial overlap, no overlap). We selected our frequencies based on values derived from Kim et al. [21]. We recruited six sighted participants (ages 20–32, four female) via convenience sampling for this study. In general, these participants had little experience with wearable technology: three had limited prior experience and two had none. As participants were not expected to be familiar with Nemeth braille, participants were given a cheat sheet to refer to during the study when identifying tactons. We acknowledge that the lack of BVI participants in this phase of the study is a limitation of the study; however we do validate the design choices made based on the results of this study with a blind individual in a subsequent study.

The perception study consisted of two phases. In the first phase, we tested participants' ability to recognize five randomly generated mathematical symbols. In the second phase, we tested participants' ability to recognize ten simple algebraic equations. These phases were conducted for each of the three locations with a random vibration and overlap pattern selected for each stimulus. In total, we asked each participant to identify 15 symbols and 30 algebraic equations. Across the participants, we employed Latin square counterbalancing to explore all design combinations an equal number of times. Each participant experienced every combination of vibrotactile pattern and frequency with repeats. At the completion of the study, participants were asked to complete a NASA-TLX survey to measure the mental workload of using this system on six metrics: mental demand, physical demand, temporal demand, performance, effort, and frustration. We used the raw version without subscale weighting and oriented the scales so that lower scores are better for all dimensions, i.e., the performance subscale marks a score of 0 as "perfect" and 20 as "failure".

The all-or-nothing accuracy for recognizing symbols and equations across the six participants is given in Fig. 3a. The posterior distal location consistently had the highest performance while the proximal location had the worst. The accuracy by vibrotactile pattern and frequency is given in Fig. 3b. The complete overlap pattern was the hardest for participants to distinguish while the no overlap pattern generally had the highest accuracy. The 10 Hz frequency had notably worse accuracy than 20 Hz or 30 Hz. On average, 20 Hz and 30 Hz had similar performance, but 20 Hz with no overlap had the best accuracy overall at 87%.

The NASA-TLX results are given in Fig. 4. Participants answered consistently across the locations despite the difference in accuracy. Except for effort, participants rated each metric on the lower side with physical demand being notably minimal. However, effort was rated high, indicating that the participants needed to concentrate. From this study, we conclude that using haptic vibrations at 20 Hz with no overlap in the posterior distal position is the clearest pattern to use in the proof of concept system.

3.2 Navigation Study

To design intuitive navigation functions, we conducted a survey with four visually impaired individuals (ages 36–42, 1 female). Each

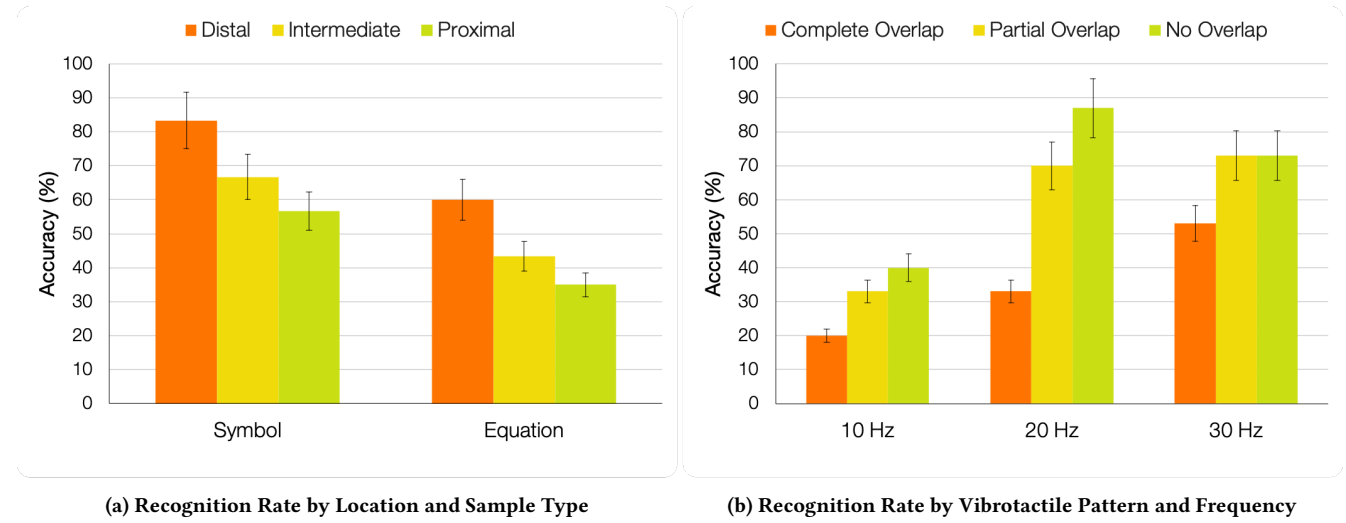


Figure 3: Recognition Rate Across Design Parameters

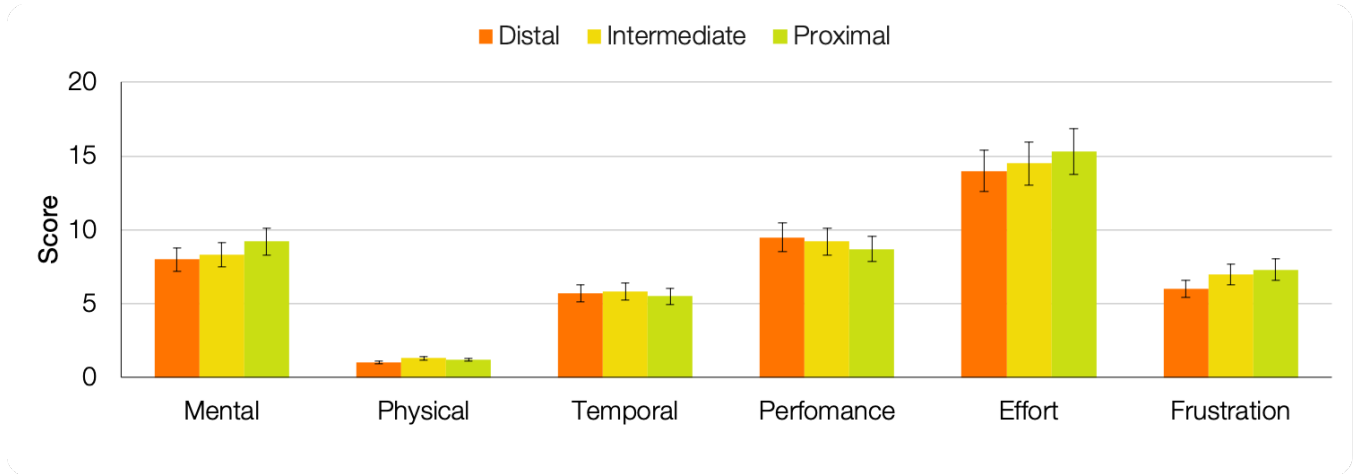


Figure 4: NASA-TLX Workload Assessments by Location (Lower is Better)

question had the same multiple choice options: Swipe Motion, Finger Input Combination, Closing Motion or Button on Palm, and an option to write in their own answer. Each question had a follow-up where participants could leave comments or any explanation. The results of the survey are given in Fig. 5. In general, there was no consensus amongst the responses, with some participants preferring swiping and others preferring multiple finger inputs for functions involving moving the “cursor”. Overall, participants preferred multiple finger inputs for changing scale (i.e., switching focus from line to symbol). We generalized this preference to all navigation functions as a design choice to be consistent.

3.3 Final Glove Design

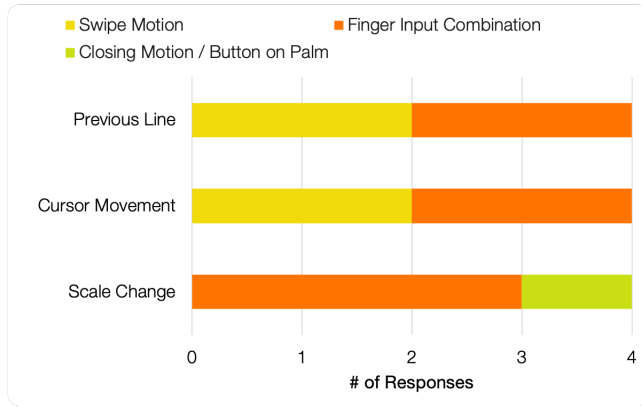
Based on the results of the perception study, we placed the vibration motors on the distal segments of each finger and thumb. For navigating through the equation, input buttons were placed on the

pads of the pinky fingers and on the outside edge of the thumbs. The final prototype of the gloves can be seen in Figure 2b. The buttons and motors for each hand were controlled through the use of individual Lilypad Arduinos.

The navigation commands we developed are shown in Table 1. For this prototype, to communicate a symbol that is comprised of multiple dots within the cell or a navigation command that consisted of multiple buttons, users were required to enter each of the input signals for the dots sequentially within a timeout window. As an example, to input in the number 0 which consists of the 3rd, 5th, and 6th dots in the braille cell, users would press the button on the left ring finger (L3), the button on the right middle finger (R2), and the button on the right ring finger (R3) in sequential order.

Table 1: Navigation Commands

Command	Button Combination
New Line Entry	Left Pinky (L4)
Backspace (by single character)	Right Pinky (R4)
Move to previous unit (line or character)	Left Ring Ringer (L3), Left Pinky (L4)
Move to next unit (line or character)	Right Ring Finger (R3), Right Pinky (R4)
Change to smaller format of navigation and read in values (change from line format to character format)	Left Thumb (L0), Left Pinky (L4)
Change to larger format of navigation and read in values (change from character format to line format)	Right Thumb (R0), Right Pinky (R4)

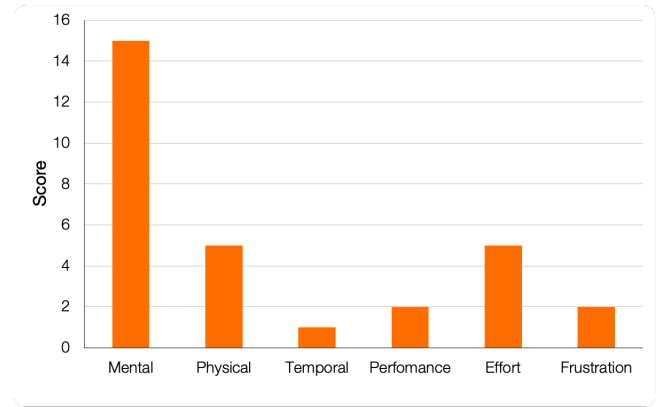
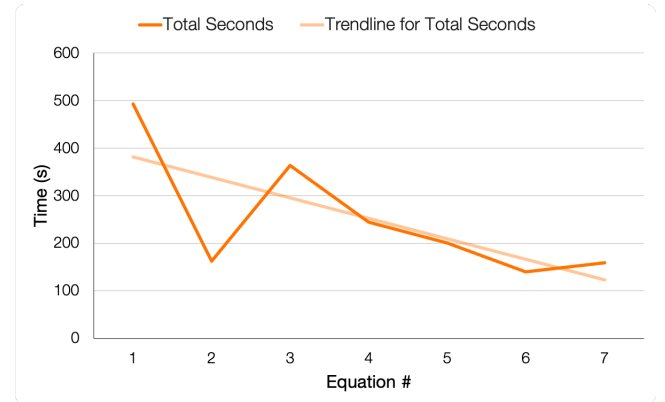
**Figure 5: Responses to Navigation Function Survey**

4 PILOT STUDY

We conducted a pilot study to determine the mental effort required to use the haptic glove interface. At the beginning of the study participants went through a training period in which they were allowed to familiarize themselves with the glove and experience the vibrations for each of the Nemeth braille patterns that could be communicated via the glove. This training period lasted 10-15 minutes. During the testing stage of the study, the user was given seven simple algebraic problems (e.g., $7 = X + 3$, $X + 5 = 3 \times 2$, $7 \times X = 14$) to solve using the gloves and user interface. For each problem, we recorded the time it took the user to complete the problem and the accuracy of the result. We did not award partial credit when evaluating the accuracy of the solutions. After completing the problems, participants were asked to complete a NASA-TLX survey to measure the mental workload of using the system.

We recruited one blind participant who was blind from birth and was very familiar with wearable technologies. She is an author on this work for her input during the ideation phase of this work. However, she was not involved in the design of the haptic glove interface or the navigation interactions, making her eligible to give feedback in the pilot study as a user.

The blind participant's experience with Nemeth braille enabled her to immediately recognize the haptic patterns. As she went through the equations, the participant stated what she was perceiving and was correct most of the time. However, she confused a symbol on three of the seven problems, e.g., + and / which use the

**(a) NASA-TLX Results (Lower is Better)****(b) Algebra Equation Answering Times****Figure 6: Results of Pilot Study**

same dots in the cell, causing her to give the wrong final answer. Over the course, her answering time per problem decreased over time, suggesting that she became adjusted to the interface over time. After completing all seven problems, she completed the NASA-TLX survey. She commented that she had to concentrate, rating mental demand high. Otherwise, she rated each subscale low. These results are given in Fig. 6.

From this study, we concluded that the design of the haptic glove was promising. Participants were able to successfully answer

algebra questions, improved over the course of the study, and rated the workload required as measured by NASA-TLX relatively low. We discuss feedback on how to improve the design of the haptic gloves in the following section.

5 DISCUSSION AND FUTURE WORK

5.1 Feedback on the Existing System

The blind participant from the pilot study provided thorough feedback about how to improve the prototype. Overall, she thought the system was intuitive and that it had excellent potential. As corroborated by her NASA-TLX results, she did not find the system difficult or frustrating to use, but she commented that she had to pay attention. A common theme of her suggestions was the need for more direct control over the tactile communication:

“Like sometimes I just wanted to mash on something that says ‘repeat the character I just felt’ because I’m not sure that I felt that.”

At multiple times during the study, she had wanted to repeat just the last symbol while in the line scale, but the system had already returned to the beginning of the line before she could switch into the symbol scale. The ability to scroll with thumbs was suggested to improve the navigation features to move between characters and lines, depending on the active mode. Discussing her experience using screen readers and the digital equation notation, MathML:

“Navigating text is fine: I can do character, word, sentence, paragraph, line, anything I want. But navigating through math equations is different, because how do you parse it? Do you jump to the next plus or the next space? There’s this kind of tree structure that they do in certain equations like in MathML”

There were some hardware limitations that need to be addressed in future iterations. For example, simultaneous buttons are not supported by the system; however the participant did make the mistake of trying to enter in inputs by pressing buttons simultaneously. For example, a common error during the study was pressing the “erase” input (R4) instead of the “move forward” input (R3 R4) because both fingers were pressed together. This can be addressed by changing the “erase” command to a combination that is further from other inputs to avoid this problem. When entering symbols, sometimes the participant would go faster than the system expected, so her intended symbols would not be correctly recorded, e.g., $X =$ became just $=$. This issue can be addressed by improving the input feedback, e.g., vibrating the finger with successful input, and having a function to confirm entry instead of a timeout. The participant thought simultaneous and sequential button presses both made sense as the former is similar to a braille while the latter is similar to a keyboard. However, she emphasized that flexibility in the input style is important, so both options will be supported in the updated hardware.

Instead of gloves, the blind participant shared the idea of using a rubbery mat with the vibrotactile feedback below the hands. During the study, she had adopted a claw-style posture to keep her fingers apart and pressed against the table to better feel the haptic feedback.

A non-wearable solution would keep the hands unrestricted and afford other interactions.

5.2 Limitations and Next Steps

The biggest limitation of our work is that only one blind participant evaluated the feasibility of the final system. Future work should not only include more visually impaired users but also look at the learnability and usability of the system over multiple uses and extended use periods. While the reported mental workload was high during the pilot study, we believe this outcome is in part due to the study being the participant’s first exposure to this interface. Our results in this work show that users across the board were able to interpret the signals faster as the study progressed, and we hypothesize that users would get faster and more accurate with more practice. With improvements to the system based off the feedback from the pilot study, this haptic interface has the potential to make algebra more accessible to the blind and visually impaired.

6 CONCLUSION

Our goal was to develop a system that aids BVI individuals in understanding, navigating, and entering in solutions to simple algebraic problems. To achieve this, we investigated the design of haptic gloves to communicate math notation through Nemeth braille mappings as well as custom navigation interactions. Preliminary results with respect to the participants’ mental workload and feedback about the design indicated that haptics are a promising medium for communicating mathematical concepts. Notably, a blind participant found the Nemeth braille mappings immediately intuitive and rapidly became accustomed to the haptic feedback.

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